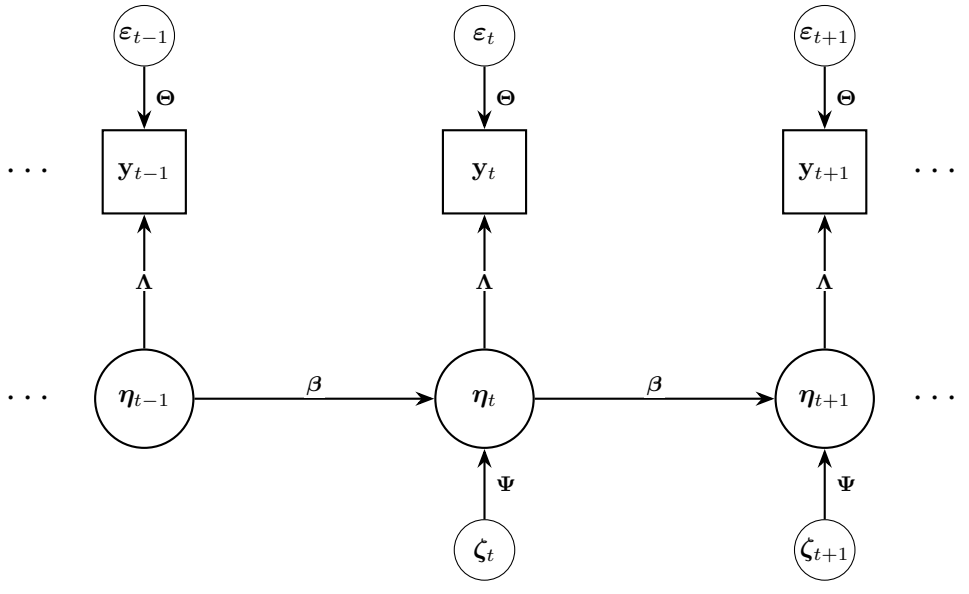


$$\boldsymbol{\beta} = \exp\left(\boldsymbol{\Phi}\Delta t\right),\ \boldsymbol{\Phi} = \begin{pmatrix} \phi_{11} & 0 & 0 \\ \phi_{21} & \phi_{22} & 0 \\ \phi_{31} & \phi_{32} & \phi_{33} \end{pmatrix} \qquad \boldsymbol{\Lambda} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\boldsymbol{\Psi} = \int_0^{\Delta t} \exp\left(\boldsymbol{\Phi}s\right) \boldsymbol{\Sigma} \exp\left(\boldsymbol{\Phi}'s\right) \mathrm{d}s,\ \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{pmatrix} \qquad \boldsymbol{\Theta} = \begin{pmatrix} \theta_{11} & 0 & 0 \\ 0 & \theta_{22} & 0 \\ 0 & 0 & \theta_{33} \end{pmatrix}$$

$\boldsymbol{\eta}_t = (\eta_1, \eta_2, \eta_3)'$ (latent), $\quad \mathbf{y}_t = (y_1, y_2, y_3)'$ (observed), $\quad \Delta t$: the interval between occasions